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Spain 02/06/1993

Rubén GONZÁLEZ

ORCID: 0000-0001-9588-1127

ru.gonzalez@uvigo.gal | https://gonzalezrvirus.com

I investigate virus evolution and host-pathogen interactions across diverse biological systems, with particular emphasis on how environmental factors and microbial communities shape infection outcomes. Using plant (*Arabidopsis thaliana*) and animal (nematode *Caenorhabditis elegans* and fly *Drosophila melanogaster*) model systems, my research integrates experimental evolution, molecular virology, and systems biology approaches to address three fundamental questions:

(i) How do different factors drive viral evolution and alter virus-host relationships, including transitions from parasitism to mutualism?

(ii) How do host genetics, viral strain variation, environmental conditions, nanomaterials, and host-associated microbial communities modulate viral pathology and infection dynamics?

(iii) What are the physiological consequences of viral infections on host fitness, development, and aging?

Education

04/2017 – 11/2021 **Ph.D. Biotechnology** w/ Santiago F. Elena
Polytechnic University of Valencia (Valencia, Spain). Honours: *cum laude*
Visiting PhD student w/ Marie-Anne Félix (01/2021 – 04/2021)
IBENS, Institut de Biologie de l'École normale supérieure (Paris, France).

09/2015 – 01/2017 **M.S. Plant Molecular and Cellular Biotechnology** w/ Santiago F. Elena
Polytechnic University of Valencia (Valencia, Spain)

09/2011 – 07/2015 **B.S. Biology** University of Vigo (Vigo, Spain)
Exchange student University of Coimbra (Coimbra, Portugal) (09/2014 – 01/2015)

Research experience

07/2026 – ongoing **Postdoctoral researcher** w/ Miguel Correa, María Mayán, David Posada
Center for Research in Nanomaterials and Biomedicine (Vigo, Spain)

01/2024 – 06/2026 **Postdoctoral researcher** w/ Carla Saleh
Institut Pasteur (Paris, France)

11/2021 – 11/2023 **Postdoctoral researcher** w/ Marie-Anne Félix
Institut de Biologie de l'École Normale Supérieure (Paris, France)

04/2017 – 09/2021 **Predoctoral researcher** w/ Santiago F. Elena

09/2018 – 09/2021 Institute for Integrative Systems Biology (Paterna, Spain)

04/2017 – 09/2018 Institute for Plant Molecular and Cell Biology (Valencia, Spain)

05/2016 – 12/2016 **Research Internship** ADM Biopolis (Valencia, Spain)

01/2015 – 07/2015 **Department research assistant** University of Vigo (Vigo, Spain)

06/2013 – 07/2013 **Volunteer Research Assistant** University of Vigo (Vigo, Spain)

Funding and awards

Year	Award	Details	Quantity
2026	Research grant	**Marie Skłodowska-Curie Actions** COFUND NANO4TALENT Postdoctoral Fellowship	168 000 €
2026	Research grant	**Pasteur Seed Grant**	10 000 €
2024	Research grant	**Pasteur-Roux-Cantarini** Postdoctoral Fellowship	2-year salary
2023	Seal of Excellence	**Marie Skłodowska-Curie Actions** Postdoctoral Fellowship	–
2021	Research grant	**European Molecular Biology Organization** Postdoctoral Fellowship (ALTF 311-2021)	143 000 €
2021	Research grant	**European Space Agency** Co-lead author: Santiago F Elena	180 682 €
2020	Exchange grant	**European Molecular Biology Organization** Short-Term Fellowship	4 450 €
2020	Exchange grant	**FISABIO-Valencia University** Chair for financing research stays	4 000 €
2020	Honourable mention	**GBE** Best Student paper published in 2019 Awards	500 €
2020	Membership	**AAAS/Science** Program for Excellence in Science	–
2020	Travel Grant	**Wellcome Genome Campus** Conference bursary	170 €
2017	Research grant	**MINECO-FEDER** FPI Scholarship for Research Training (BES-2016-077078)	92 750 €
2015	Research grant	**MINECO-FEDER** Department collaboration (BOE-A-2015-700)	2 000 €
2014	Exchange grant	**Xunta de Galicia** ERASMUS+ exchange grant complement	1 100 €
2014	Exchange grant	**European Commission** ERASMUS+ Fellowship	1 700 €

Service

>EDITORIAL AND PEER REVIEW

Associate Editor

Virology Journal (2026-ongoing)

Editorial Board Member

npj Viruses (2023-ongoing)

Reviewer

Peer Community in Evolutionary Biology, *Journal of General Virology*, *Journal of Virology*, *Plants*, *PLoS Pathogens, Evolution Letters*, *Virology*, *BMC Biology, Nature Communications, Journal of Evolutionary Biology, npj Viruses*.

Grant evaluation committee

ANR - AAPG 2024 (panel "CE15 – Immunologie, Infectiologie et Inflammation"), UK Research and Innovation Funding Service (panel OPP490: BBSRC Responsive Mode: 24RM2)

External examiner

For the PhD thesis of Joan Márquez Molins (2022) and Adrián Pedreira Bar (2024).

>REPRESENTATION AND ORGANIZATION

Postdoc association representative of IBENS (2022-2023) and Pasteur (2024-2026).

Organization team member Institut Pasteur departmental Virology days (2024 and 2026), Paris Postdocs Seminar (2024), IBENS Ecology and Evolution Biology department Seminars (2022-2023).

Scientific Committee member of the 2nd PhD Meeting in Plant Science.

Session chair Institut Pasteur Journées Départementales de Virologie (2024), EMBO Fellows Meeting (2025), GRS Biology of Aging (2025), Jaques Monod Conference: Insect immunity (2026).

Invited talks

12/2025 **When Viruses Stay: How Persistent Infections Drive Viral Evolution and Shape Host Biology.** Ecology and Evolutionary Biology cluster seminar series University of Sheffield (Sheffield, UK).

04/2023 **Monobacterial environments regulate *Caenorhabditis elegans* resistance to viral infection.** Verena Institute seminar series ONLINE (Washington, USA).

01/2023 **Impact of host developmental stage on virus evolution.** Keynote talk at French National Plant Virus Symposium (Aussois, France).

03/2021 **A plant virus evolves under drought conditions to promote host survival.** Department seminar at CMESS (Vienna, Austria).

07/2020 **Virus evolution under drought conditions.** Ecology and Evolution of Infectious Disease Seminar, UC Berkeley (Berkeley, USA).

Supervision

PhD student

2020-2024 Izan Melero. *Co-supervised with Santiago Elena.*
(University of Valencia, Spain) *MCIU Fellow @ I2SysBio*

PhD student [Scientific supervision]

2023-2024 Mauro Castelló-Sanjuan. *Thesis director: Carla Saleh*
(Université Paris Cité, France) *@ Institut Pasteur*

PhD student [Internship supervision]

08/09-2024 Hiroko Kobayashi (Hokkaido University, Japan) *WISE Fellow @ Institut Pasteur*

04/06-2026 Esmeralda García Legarda (University of Valencia, Spain) *CSIC Fellow @ Institut Pasteur*

Research internships

06-12/2018 Javier Martínez-Latorre *JAE-Intro Fellow @ I2SysBio*

02-04/2019 Izan Melero *ADEIT/Universitat de València Practicum @ I2SysBio*

05-07/2019 Enric Pérez *ADEIT/Universitat de València Practicum @ I2SysBio*

Publications

* co-authorship † corresponding author

1. **González R†**, Castelló-Sanjuán M, Saleh MC† (2026) Enteric viral infections promote systemic accelerated aging in *Drosophila*. Science Advances 12,eaec1735
2. Villena-Giménez A*, Legarda EG*, **González R**, Castiglioni VG, Elena SF† (2026) Cosmic silence and viral noise: transcriptomic crosstalk in *Caenorhabditis elegans* under simulated space conditions. Frontiers in Microbiology 17:1781245

3. Villena-Giménez A, Castiglioni VG, Legarda EG, Muñoz-Sánchez JC, **González R**, Elena SF† (2026) Reduced gravity and muon flux absence affect *Caenorhabditis elegans* life history traits and viral infection. Microbiology Spectrum 14:e03580-25
4. **González R†**, Castelló-Sanjuán M, Romoli O, Blanc H, Kobayashi H, Nigg J, Saleh MC† (2025) Microbiome composition modulates the lethal outcome of Drosophila A virus infection. Cellular and Molecular Life Sciences 83:61
5. Castelló-Sanjuán M, **González R**, Bergman A, Blanc H, Frangeul L, Nigg J, Saleh MC† (2025) The pupal stage is a developmental window for RNA virus persistence in *Drosophila*. bioRxiv [preprint] 2025.11.07.687216
6. Melero I†, Castiglioni VG, Olmo-Uceda MJ, Olmedo M, **González R†**, Elena SF† (2025) The interplay between *Caenorhabditis elegans* larval development and Orsay virus infection. iScience 28:114193.
7. Castelló-Sanjuán M, **González R**, Romoli O, Blanc H, Nigg J, Saleh MC† (2025) Persistent viral infections impact key biological traits in *Drosophila melanogaster*. PLoS Biology 23:e3003437
8. **González R***, Koh C*, Virlon B*, Merkling SH†, Vartanian JP † (2025) A look back at the departmental Virology Days of the Institut Pasteur (Le Touquet, May 13-15, 2024). Journal of Molecular Cell Biology mjae052
9. Melero I, Gómez-Cadenas A, **González R†**, Elena SF† (2024). Transcriptional and hormonal profiling uncovers the interactions between plant developmental stages and RNA virus infection. Journal of General Virology. 105, 002023.
10. **González R†**, Félix M.-A.† (2024) Natural monobacterial environments modulate viral infection in *Caenorhabditis elegans*. PLoS Pathogens. 20, e1011947,
11. **González R†**, Félix M.-A. (2024) *Caenorhabditis elegans* immune responses to intracellular pathogens. Developmental & Comparative Immunology 154, 105148.
12. **González R†**, Félix M.-A†. (2024) *Caenorhabditis elegans* defective-pharynx and constipated mutants are resistant to Orsay virus infection. microPublication Biology, 2024: 10.17912/micropub.biology.001166.
13. Castiglioni VG, Olmo-Uceda MJ, Martín S, Félix M.-A, **González R†**, Elena SF† (2024) Experimental evolution of an RNA virus in *Caenorhabditis elegans*. Infection, Genetics and Evolution. 123, 105623
14. Melero I, Elena SF, **González R†** (2023) Viral strain-dependent impact of plant developmental stages on the nestedness and modularity of plant-virus interaction matrices. microPublication Biology. 10.17912/micropub.biology.000943.
15. Butkovic A*, Ellis TJ*, **González R***, Jaegle B, Nordborg M†, Elena SF† (2023) Genetic basis of *Arabidopsis thaliana* responses to infection by naïve and adapted isolates of turnip mosaic virus. eLife, 12:RP89749
16. Melero I, **González R†**, Elena SF† (2023) Host developmental stages shape the evolution of a plant RNA virus. Philosophical Transactions of the Royal Society B. 378:20220005.
17. Butkovic A*†, **González R*†** (2022) A brief view on factors that affect plant virus evolution. Frontiers in Virology. 2:994057.
18. Navarro R, Ambrós S, Butković A, Carrasco JL, **González R**, Martínez F, Wu B, Elena SF†. (2022) Defects in plant immunity modulate the rates and patterns of RNA virus evolution. Virus Evolution, veac059.

19. **González R†,** Elena SF. (2021) The interplay between the host microbiome and pathogenic viral infections. mBio. 12: e02496-21.
20. **González R**, Butković A, Escaray F, Martínez-Latorre J, Melero I, Pérez-Parets E, Gómez-Cadenas A, Carrasco P, Elena SF†. (2021) Plant virus evolution under strong drought conditions results in a transition from parasitism to mutualism. PNAS. 118, e2020990118.
    >> Commentary: Dolja, VV†. (2021) Rapid emergence of virus–host mutualism under stress. PNAS. 118, e2100936118.
21. Butković A, **González R**, Elena SF†. (2021) Revisiting *Orthotospovirus* phylogeny using full-genome data and testing the contribution of selection, recombination and segments reassortment in the origin of new species. Archives of Virology. 166, 491–499.
22. **González R*†**, Butković A*, Elena SF†. (2020) From foes to friends: viral infections expand the limits of host phenotypic plasticity. Advances in Virus Research, 106: 85-121.
23. Butković A, **González R**, Rivarez MPS, Elena SF†. (2020) A genome-wide association study identifies Arabidopsis thaliana genes that contribute to differences in the outcome of infection with two Turnip mosaic potyvirus strains that differ in their evolutionary history and degree of host specialization. Virus Evolution, veab063.
24. **González R†**, Butković A, Rivarez MPS, Elena SF. (2020) Natural variation in Arabidopsis thaliana rosette area unveils new genes involved in plant development. Scientific Reports, 10,17600.
25. Butkovic A, **González R**, Cobo I, Elena SF†. (2020) Adaptation of turnip mosaic potyvirus to a specific niche reduces its genetic and environmental robustness Virus Evolution, 6: veaa041
26. **González R**, Butković A, Elena SF†. (2019) Role of host genetic diversity for susceptibility-to-infection in the evolution of virulence of a plant virus. Virus Evolution, 5: vez024.
27. **González R*,** Wu B*, Li X*, Martínez F†, Elena SF†. (2019) Mutagenesis scanning uncovers evolutionary constraints on tobacco etch potyvirus membrane-associated 6K2 protein. Genome Biology and Evolution, 11: 1207–1222.
28. Rolán-Alvarez† E, Carvajal-Rodríguez A, de Coo A, Cortés B, Estévez D, Ferreira M, **González R**, Briscoe AD. (2015) The scale-of-choice effect and how estimates of assortative mating in the wild can be biased due to heterogeneous samples. Evolution, 69: 1845–1857.

Conferences and Meetings

- Jaques Monod Conferences: Insect immunity, the guardian of homeostasis and symbiosis (Roscoff, France) Enteric viral infections promote systemic accelerated aging in *Drosophila*. (2026) Talk.
- GRS and GRC Biology of Aging (Lucca, Italy) Enteric viral infections promote systemic accelerated aging in *Drosophila*. (2025) Poster.
- 4th Colloquium Aging in the web of life (Paris, France) (2025). Attended.
- Late-life Diseases Symposium (Paris, France) Enteric viral infections promote systemic accelerated aging in *Drosophila*. (2025) Poster.

- EMBO Fellows Meeting (Heidelberg, Germany) Viral infections reduce *Drosophila* lifespan through accelerated aging. (2025) Talk.
- LABEX IBEID International Conference (Paris, France) Bacteriome composition modulates viral infections in *Drosophila melanogaster* (2024). Poster
- Institut Pasteur *Journées Départementales de Virologie* (Le Touquet, France) Bacteriome modulation of viral infections in *Drosophila melanogaster*. (2024). Talk
- 24th International C. elegans Conference (Glasgow, UK) *Caenorhabditis elegans* susceptibility to virus infection is modulated by its naturally associated bacteria. (2023). Talk
- EMBL - The organism and its environment (Heidelberg, Germany) *Caenorhabditis elegans* susceptibility to virus infection is modulated by its naturally associated bacteria. (2023). Talk
- VerMidi XXIV (Paris, France) *Caenorhabditis elegans* susceptibility to virus infection is modulated by its naturally associated bacteria. (2023). Talk
- VIP&DIF (Paris, France) (2023) Natural monobacterial environments modulate viral infection in *Caenorhabditis elegans*. Talk and Poster
- Evolutionary Biology of Caenorhabditis and Other Nematodes (Hamilton, Canada) Sensitivity of *C. elegans* to Orsay virus is suppressed by some bacteria and by a *haao-1* reduction-of-function polymorphism. (2022). Talk.
- Wellcome Genome Campus Evolutionary Systems Biology (2022). Attended.
- Wellcome Genome Campus Virus Genomics and Evolution (Hixton, UK) Plant aging increases susceptibility to Turnip mosaic potyvirus infection and influences virus evolution. (2021). Poster.
- Wellcome Genome Campus Single Cell Biology (2020). Attended.
- Wellcome Genome Campus 3rd meeting on Evolutionary Systems Biology (Hixton, UK) Virus evolution in the pathosystem turnip mosaic potyvirus – *Arabidopsis thaliana* under drought conditions selects for viruses who enhance the host tolerance to water deficiency. (2020). Poster.
- Experimental Evolution Workshop (Biar, Spain) Evolving towards mutualism: an experimental study on the virus-host interactions changes induced by the environment. (2019). Talk.
- 10th Workshop in Virus Evolution (State College Pennsylvania, USA) Evolving in drought conditions: consequences in the interaction between a potyvirus and *Arabidopsis thaliana*. (2019). Poster.
- 24th European Meeting of PhD Students in Evolutionary Biology (Bubión, España) Evolution of RNA viruses in hosts with variable susceptibility. (2018). Talk.
- VI Spanish Worm Meeting (Valencia, Spain) Evaluation of *Caenorhabditis elegans* as experimental host for a thrip-transmitted plant RNA virus. (2017). Talk.

Courses

2026 **Equity and Bias-Free Scientific Practice**
CINBIO
2024 **EMBO Laboratory Leadership for Postdocs** EMBO Solutions GmbH
2021 **Research Integrity** Epigeum

2021 **YIP PhD course** EMBO Solutions GmbH

2020 **Program certificate in Data Analysis for Life Science** HarvardX

2017 **Ethics in research** Polytechnic University of Valencia (UPV)

2017 **Statistics and mathematical methods for Research** Polytechnic University of Valencia

2017 **Rewriting science: Avoiding mistakes in written and spoken scientific English** UPV